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Control Your AI Mind

Applied Lab Book

Corpus date: 2026-06-10

This lab book converts the source corpus into hands-on exercises. The labs are ordered from low-risk and visible to high-risk and judgment-heavy.

General Lab Rules

1. Never modify raw data.
2. Always save scripts, not just outputs.
3. Every output must have a path and a provenance note.
4. Every lab ends with a verification memo.
5. If the agent is blocked, document the blocker rather than silently changing scope.

Lab 0. Environment And Repo Contract

Goal. Set up a project folder that an agent can safely work inside.

Estimated time. 60-120 minutes.

Steps. 1. Create folder tree. 2. Initialize git. 3. Write README. 4. Write AGENTS.md or CLAUDE.md. 5. Define data/privacy policy.

Starter prompt.

We are doing Lab 0: Environment And Repo Contract. Do not start coding yet. First restate the

- ↪ task contract, list the inputs you need, identify possible failure modes, and propose
- ↪ acceptance checks. After I approve, execute only the approved steps and save an audit
- ↪ note.

Acceptance checks.

- The output path is explicit.
- The source/data path is explicit.
- The agent's assumptions are written down.
- There is at least one independent sanity check.
- The final memo states what remains uncertain.

Deliverable. Repo setup contract and risk checklist

Common failure. Students let the agent choose scope. The instructor should stop them and force a task contract before execution.

Lab 1. Source Map For A Research Question

Goal. Build a source atlas before asking the agent to write anything.

Estimated time. 60-120 minutes.

Steps. 1. Define one research question. 2. List source types. 3. Crawl or collect metadata. 4. Label source credibility. 5. Write public/private policy.

Starter prompt.

We are doing Lab 1: Source Map For A Research Question. Do not start coding yet. First

- ↪ restate the task contract, list the inputs you need, identify possible failure modes, and
- ↪ propose acceptance checks. After I approve, execute only the approved steps and save an
- ↪ audit note.

Acceptance checks.

- The output path is explicit.
- The source/data path is explicit.
- The agent's assumptions are written down.
- There is at least one independent sanity check.
- The final memo states what remains uncertain.

Deliverable. Source map table

Common failure. Students let the agent choose scope. The instructor should stop them and force a task contract before execution.

Lab 2. Empty Folder To Figure

Goal. Replicate the Paul-style descriptive workflow from a blank directory.

Estimated time. 60-120 minutes.

Steps. 1. State descriptive question. 2. Find public data. 3. Download raw data. 4. Write processing script. 5. Generate one figure. 6. Write reproducibility note.

Starter prompt.

We are doing Lab 2: Empty Folder To Figure. Do not start coding yet. First restate the task

- ↪ contract, list the inputs you need, identify possible failure modes, and propose
- ↪ acceptance checks. After I approve, execute only the approved steps and save an audit
- ↪ note.

Acceptance checks.

- The output path is explicit.
- The source/data path is explicit.
- The agent's assumptions are written down.
- There is at least one independent sanity check.
- The final memo states what remains uncertain.

Deliverable. Figure plus reproducibility note

Common failure. Students let the agent choose scope. The instructor should stop them and force a task contract before execution.

Lab 3. Scrape To Structured Database

Goal. Turn unstructured web/source text into a queryable database.

Estimated time. 60-120 minutes.

Steps. 1. Define sample. 2. Download/cache raw pages. 3. Parse content. 4. Design schema. 5. Store in DuckDB/SQLite. 6. Run quality report.

Starter prompt.

We are doing Lab 3: Scrape To Structured Database. Do not start coding yet. First restate the

- ↪ task contract, list the inputs you need, identify possible failure modes, and propose
- ↪ acceptance checks. After I approve, execute only the approved steps and save an audit
- ↪ note.

Acceptance checks.

- The output path is explicit.
- The source/data path is explicit.
- The agent's assumptions are written down.
- There is at least one independent sanity check.
- The final memo states what remains uncertain.

Deliverable. Database file plus quality report

Common failure. Students let the agent choose scope. The instructor should stop them and force a task contract before execution.

Lab 4. Large Data With Parquet And DuckDB

Goal. Move from flat files to scalable, documented data infrastructure.

Estimated time. 60-120 minutes.

Steps. 1. Inspect file size and columns. 2. Convert CSV to parquet. 3. Create DuckDB database. 4. Create metadata table. 5. Build aggregate table. 6. Run sanity checks.

Starter prompt.

We are doing Lab 4: Large Data With Parquet And DuckDB. Do not start coding yet. First

- ↪ restate the task contract, list the inputs you need, identify possible failure modes, and
- ↪ propose acceptance checks. After I approve, execute only the approved steps and save an
- ↪ audit note.

Acceptance checks.

- The output path is explicit.
- The source/data path is explicit.
- The agent's assumptions are written down.
- There is at least one independent sanity check.
- The final memo states what remains uncertain.

Deliverable. DuckDB/parquet workflow plan or prototype

Common failure. Students let the agent choose scope. The instructor should stop them and force a task contract before execution.

Lab 5. Writing As Thinking

Goal. Use AI as editor without surrendering argument ownership.

Estimated time. 60-120 minutes.

Steps. 1. Pick result paragraph. 2. Label each sentence. 3. Tie claims to evidence. 4. Ask for comments only. 5. Rewrite manually. 6. Document changes.

Starter prompt.

We are doing Lab 5: Writing As Thinking. Do not start coding yet. First restate the task

- ↪ contract, list the inputs you need, identify possible failure modes, and propose
- ↪ acceptance checks. After I approve, execute only the approved steps and save an audit
- ↪ note.

Acceptance checks.

- The output path is explicit.
- The source/data path is explicit.
- The agent's assumptions are written down.

- There is at least one independent sanity check.
- The final memo states what remains uncertain.

Deliverable. Claim audit and revised paragraph

Common failure. Students let the agent choose scope. The instructor should stop them and force a task contract before execution.

Lab 6. Build A Research Skill

Goal. Encode a repeated workflow into a reusable instruction file.

Estimated time. 60-120 minutes.

Steps. 1. Choose recurring task. 2. Define trigger. 3. Define input/output. 4. Write process. 5. Add checks. 6. Run two tests.

Starter prompt.

We are doing Lab 6: Build A Research Skill. Do not start coding yet. First restate the task

- ↪ contract, list the inputs you need, identify possible failure modes, and propose
- ↪ acceptance checks. After I approve, execute only the approved steps and save an audit
- ↪ note.

Acceptance checks.

- The output path is explicit.
- The source/data path is explicit.
- The agent's assumptions are written down.
- There is at least one independent sanity check.
- The final memo states what remains uncertain.

Deliverable. SKILL.md-style artifact

Common failure. Students let the agent choose scope. The instructor should stop them and force a task contract before execution.

Lab 7. Applied Econometrics Verification

Goal. Build a Scott-style checklist for a DiD/CSDID or panel workflow.

Estimated time. 60-120 minutes.

Steps. 1. Define estimand. 2. Build cohort table. 3. Plot rollout. 4. Check covariates. 5. Plot outcome paths. 6. Check package versions. 7. Write memo.

Starter prompt.

We are doing Lab 7: Applied Econometrics Verification. Do not start coding yet. First restate

- ↪ the task contract, list the inputs you need, identify possible failure modes, and propose
- ↪ acceptance checks. After I approve, execute only the approved steps and save an audit
- ↪ note.

Acceptance checks.

- The output path is explicit.
- The source/data path is explicit.
- The agent's assumptions are written down.
- There is at least one independent sanity check.
- The final memo states what remains uncertain.

Deliverable. Verification checklist and memo

Common failure. Students let the agent choose scope. The instructor should stop them and force a task contract before execution.

Lab 8. Research-Agent OS Capstone

Goal. Assemble one controlled workflow from question to verified output.

Estimated time. 60-120 minutes.

Steps. 1. Create source map. 2. Write task contract. 3. Run agent workflow. 4. Produce output. 5. Run adversarial review. 6. Write final audit.

Starter prompt.

We are doing Lab 8: Research-Agent OS Capstone. Do not start coding yet. First restate the

- ↪ task contract, list the inputs you need, identify possible failure modes, and propose
- ↪ acceptance checks. After I approve, execute only the approved steps and save an audit
- ↪ note.

Acceptance checks.

- The output path is explicit.
- The source/data path is explicit.
- The agent's assumptions are written down.
- There is at least one independent sanity check.
- The final memo states what remains uncertain.

Deliverable. Capstone OS map plus output artifact

Common failure. Students let the agent choose scope. The instructor should stop them and force a task contract before execution.

Capstone Package Template

```
project/
  README.md
  AGENTS.md or CLAUDE.md
  sources/
    source_map.md
  data/
    raw/
    processed/
  scripts/
  outputs/
    figures/
    tables/
  skills/
  verification/
    verification_memo.md
  final/
    final_output.md or pdf
```

Final Presentation Questions

1. What did you delegate?
2. What did you refuse to delegate?
3. What sources did the agent use?
4. What did the agent get wrong or almost get wrong?
5. How did you verify the output?
6. What would you improve before using this workflow on real research?